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ProVenTL: a transfer-learning framework for predicting peptide–protein interactions derived from snake venom for cancer therapeutics

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Abstract

Accurate prediction of peptide–protein interactions (PepPI) is crucial for advancing peptide-based anticancer drug design. In this study, we introduce ProVenTL, a computer-aided molecular design framework that leverages transfer learning and protein language model embeddings to enhance PepPI prediction accuracy and interpretability. Two complementary strategies were explored: (i) fine-tuning a CAMP model pretrained on large-scale PepPI data from the Protein Data Bank (PDB) using a curated dataset of *Calloselasma rhodostoma* venom peptides and cancer-related proteins, and (ii) integrating ProtT5 embeddings with stacked autoencoder–deep neural networks (SAE–DNN) and TabNet classifiers. Models were comprehensively benchmarked against baseline configurations and representative deep-learning approaches using standard classification metrics, while biological relevance was evaluated through functional enrichment and pathway analysis of top-ranked predictions. Compared with baseline configurations and conventional deep-learning approaches, the ProtT5-based SAE–DNN model achieved the best performance (accuracy = 0.78; ROC–AUC = 0.86), demonstrating improved generalization capability on a small, domain-specific venom peptide dataset. The model identified key targets such as TRBC2, CD274, HIF1AN, PCSK9, and PLAU, which are associated with pathways involved in immune suppression, hypoxia regulation, lipid metabolism, and metastasis. This study highlights the utility of transfer learning and protein language models for PepPI prediction in data-limited scenarios and establishes a computational framework for prioritizing snake-venom-derived peptides for anticancer drug discovery and future experimental validation.

Keywords Cancer · Deep learning · Peptide–protein interaction · Transfer learning · Venom

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Introduction

Cancer remains among the deadliest diseases worldwide. In 2022, 19.98 million new cases and 9.74 million deaths were recorded, where Asia contributed to 9.83 million cases (49.2%) [1]. The global burden is projected to reach 20.7 million cases and 12.7 million deaths by 2040, reflecting population aging [2]. These burdens underscore the requirement for more effective, safer, and affordable therapies. One active avenue is synthetic peptide-based therapeutics, which provide precise targeting, improved molecular stability, and efficient production and modification [3]. Advances in synthetic peptide–based therapies have expanded applications to peptide drug conjugates (PDCs), peptide vaccines, and tumor imaging, underscoring the requirement for innovation in peptide design and delivery to improve efficacy [4]. A comprehensive review identified that most cancer-therapy candidates to date originated from synthetic peptides [5]. The CXCR4 antagonist LY2510924 was safe up to 20 mg/day in a phase I trial but showed limited efficacy, with only 20% achieving stable disease despite increased $CD34^+$ counts [6]. Parallel strategies, such as microRNA targeting, virotherapy, and CRISPR/Cas9, also show potential in vitro and in vivo [7]. However, rapid degradation, poor membrane permeability, short half-life, limited bioavailability, and tumor and target variability continue to constrain clinical translation [8, 9]. These limitations have motivated exploration of natural-source therapeutics [10]. Natural peptides are promising owing to receptor-specific recognition and selective activation of intracellular pathways [11]. They provide high biocompatibility, sustainable production potential, and evolutionarily optimized structures, making them strong candidates for safe and effective therapies [12]. Examples include plitidepsin and marine bacteriocins, which inhibit cancer-cell proliferation with minimal toxicity [13], and plant peptides such as lunasin and RiPPs members, which induce apoptosis and inhibit the cell cycle [14, 15]. However, animal-derived peptides remain comparatively underexplored [16].

The venom of *Calloselasma rhodostoma* (C. *rhodostoma*) is a promising peptide source for cancer therapy. Key constituents include metalloproteases, disintegrins, phospholipase A_2 (PLA_2), and L-amino acid oxidases (LAAO), which show selective cytotoxicity through apoptosis induction and angiogenesis inhibition in vitro and in vivo [17]. CR-LAAO triggers HL-60 and HepG2 cell death through H_2O_2 generation, caspase activation, and BAX upregulation [18]. Transcriptomic analyses identify zinc metalloproteinases, disintegrins, and rhodocytin as therapeutic candidates [19]. The species' broad Southeast Asian distribution and Viperidae bioactive repertoire further support this therapeutic potential [20, 21]. Rhodostomin, a disintegrin, is among the best-studied compounds; it inhibits osteosarcoma-cell proliferation, tumor cell-mediated platelet aggregation, and cancer cell adhesion and metastasis [17, 22]. However, experimental identification of therapeutic peptide-cancer protein interactions remains limited owing to cost, time, and targeting complexity. Computational bioinformatics and multi-omics approaches enable more efficient candidate screening [23], but predicting protein–protein interactions, particularly those involving intrinsically disordered proteins, still challenges the design of highly specific interactions [24].

Machine learning (ML) and deep learning (DL) strategies now predict peptide-protein interactions with high accuracy. Hybrid and transformer-based architectures improve early-stage screening efficiency [25]. ML-driven automated design accelerates discovery from feature extraction through candidate selection to in silico optimization [26]. SAE-DNN achieved 0.82 accuracy, 0.48 precision, and 0.76 ROC–AUC, but generalizability and positive-interaction detection remain limited owing to small training sets and high computational demands [27]. PepCNN, combining convolutional neural networks (CNNs) with protein-language-model embeddings and structural features, improves specificity, precision, and AUC [28]. CAMP, an attention-augmented CNN model, supports binary classification and binding-residue identification, reaching up to 91% accuracy [29]. DL models for peptide–protein interactions frequently falter on generalization, data efficiency, and computational efficiency. Transfer learning (TL) offsets these limits by transferring knowledge from large source domains, boosting performance in data-scarce targets [30, 31]. TAc-based transfer outperforms from scratch training for compound classification [32]. ProtT5 captures complex sequence semantics for broad tasks [33]. TL also accelerates training and improves cross-domain generalization, and freezing or fine-tuning on multi-scale CNNs is helpful on small datasets, including protein–virus prediction [30, 34]. Hence, TL is well suited for biomolecular interaction models using underexplored sources such as snake-venom peptides.

In this study, a TL-based approach was developed for peptide–protein interaction (PepPI) prediction, focusing on natural peptides from the venom of C. *rhodostoma* with anticancer potential. The proposed model, ProtT5 venom TL (ProVenTL), leverages ProtT5 embeddings pretrained on a large-scale protein corpus and is subsequently fine-tuned for the domains of snake-venom peptides and cancer-related proteins. This design improves attention-based interpretability and training efficiency while mitigating data scarcity and feature gaps noted in prior studies. ProVenTL delivers fast, accurate, and interpretable predictions to aid therapeutic exploration of venom-derived peptides in cancer.

The main contributions of this study are as follows. First, we propose a novel approach, ProVenTL, that leverages ProtT5 embeddings to improve the accuracy and efficiency of peptide–protein interaction prediction, particularly for peptides derived from C. *rhodostoma* venom. Second, we incorporate an attention-based DL model to enable fast, biologically interpretable predictions and benchmark its performance against recent state-of-the-art approaches for PepPI. Finally, we perform functional enrichment analysis on the predicted interactions between C. *rhodostoma* peptides and cancer-related proteins to assess biological relevance and support potential therapeutic applications in cancer treatment.

Related works

Computational prediction of peptide-protein interactions (PepPI) is increasingly adopted because experimental assays are expensive and time-consuming [25]. Two principal methods are used: sequence-based, which derives predictive representations from amino acid order, and structure-based, which uses 3D information and molecular docking to model binding affinity [25]. Within the ML/DL landscape, SAE-DNN (frequently paired with random forest/XGBoost) has reported up to 0.859 accuracy, 0.663 precision, and 0.697 ROC–AUC; however, it remains limited by large training-data requirements and modest generalization [27]. TabNet effectively captures salient features from biological tabular data but faces similar challenges regarding data scale and generalization [35]. Meanwhile, CAMP, an attention-augmented CNN model, supports binary classification and binding-residue identification, achieving accuracies up to 91% in certain evaluations [29].

Recent advances have introduced specialized deep learning frameworks for peptide prediction across diverse biomedical applications. For example, DeepAIPs-SFLA integrates evolutionary and structural image-based representations with optimization strategies and convolutional networks to improve the identification of anti-inflammatory peptides [36]. Similarly, pACPs-DNN employs matrix-based evolutionary and structural encodings combined with attention-driven deep neural networks for accurate anticancer peptide prediction [37], while pACP-HybDeep integrates transformer-derived semantic features with structural descriptors and hybrid deep architectures to enhance predictive performance [38]. In addition, pNPs-CapsNet uses pretrained protein-language model embeddings and capsule neural networks for neuropeptide classification [39]. TargetCLP further adopts a multi-view feature integration strategy, combining transformed evolutionary features and ESM-based embeddings via weighted fusion and feature selection, for clathrin protein prediction [40]. In contrast, SBSM-Pro applies physicochemical grouping, sequence alignment, and multiple kernel learning with support vector machines to improve protein sequence classification performance [41]. These recent predictors demonstrate the rapid evolution of deep learning–based peptide modeling approaches; however, most focus primarily on single-sequence functional classification rather than explicit modeling of peptide–protein interaction pairs.

Interest in natural peptides is growing owing to high biocompatibility, receptor-specific recognition, prospects for sustainable production, and evolutionarily optimized structures [10–12]. The venom of C. *rhodostoma* contains metalloproteases, disintegrins, phospholipase A_2 (PLA_2), and L-amino acid oxidases (LAAO) that show selective cytotoxicity through apoptosis induction and angiogenesis inhibition in vitro and in vivo [17]. CR–LAAO triggers HL-60 and HepG2 cell death through H_2O_2 generation, caspase activation, and upregulation of BAX [18], while rhodostomin suppresses osteosarcoma proliferation, tumor cell-mediated platelet aggregation, and processes of adhesion and metastasis [22]. Transcriptomic analyses further identify zinc metalloproteinases, disintegrins, and rhodocytin as therapeutic candidates, reinforced by the broad Southeast Asian distribution of C. *rhodostoma* and the Viperidae repertoire of immunologically active and anticancer compounds [19–21].

To address data and generalization constraints in DL-based PepPI, TL transfers knowledge from large source domains to data-scarce targets and has improved performance in drug-target interaction prediction and compound classification [30–32]. In this context, ProtT5 provides rich protein-sequence embeddings and is used as a feature representation for SAE-DNN and TabNet to enhance representation quality and training efficiency on small biological datasets [28]. Freezing and fine-tuning strategies further aid cross-domain generalization, including in protein-virus interaction prediction [34]. However, TL with transformer embeddings for modeling interactions of venom-derived peptides, particularly those from C. *rhodostoma*, remains limited, presenting a gap that motivates domain-tailored PepPI frameworks [16, 30].

Materials and methods

This section outlines the systematic approach used to develop and evaluate transfer learning–based models for peptide–protein interaction prediction. The workflow visualized in Fig. 1 comprises six stages: (i) data acquisition and balanced-pair construction, including negative sampling of peptide–protein non-interactions; (ii) preprocessing to clean and harmonize sequences and labels; (iii) feature representation via a descriptor input matrix (protein descriptors) and a ProtT5 input matrix (transformer embeddings); (iv) dataset splitting and model training with CAMP, SAE-DNN, and TabNet in a transfer learning setup; (v) performance evaluation and model selection using standard classification metrics; and (vi) downstream inference and biological validation—i.e., predicting interactions for *Calloselasma rhodostoma* venom peptides followed by functional enrichment of the predicted cancer-related protein targets.

Data acquisition

This study used three primary datasets: a peptide–protein interaction dataset, cancer-related protein dataset, and C.

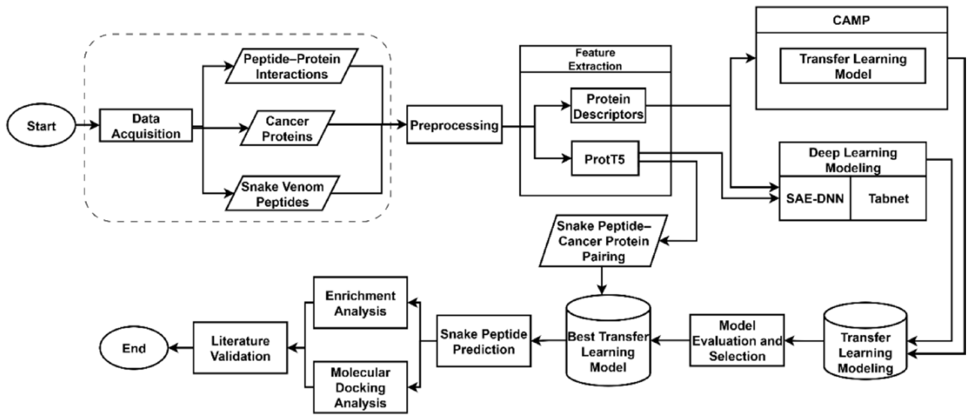


Fig. 1 Workflow of the proposed methodology

rhodostoma venom-peptide dataset; the third served as a test set to predict novel interactions between peptides and cancer-associated proteins. The peptide–protein interaction dataset was constructed by integrating four major resources: the RCSB Protein Data Bank (PDB) [42], SIFTS [43], UniProt [44], and DrugBank [45]. PDB provided 82,288 structural entries of proteins and peptides, while SIFTS enabled mapping of 764,866 residue-level entries from PDB IDs to UniProt IDs, linking structural data with biological annotations. UniProt was used to retrieve protein classifications and annotations, particularly to identify proteins involved in cancer pathways. DrugBank contributed 15,041 entries of therapeutic peptides, including known protein targets.

All protein and peptide sequences were collected and stored in FASTA format. We filtered peptides of length $\leq$ 50 residues and proteins of 50–1500 residues. The resulting sequence pairs were analyzed using the protein–ligand interaction profiler (PLIP) [46], which detects seven types of noncovalent interactions such as hydrogen bonds, hydrophobic contacts, π–stacking, π–cation interactions, salt bridges, water bridges, and halogen bonds. From this step, 37,828 pairs with structural interaction evidence were obtained. Mapping to UniProt using SIFTS yielded 37,802 structurally consistent interaction pairs. After integrating therapeutic-peptide data from DrugBank and applying filters (duplicate removal, null-value elimination, and invalid-sequence exclusion), a final set of 3,470 positive interaction pairs was curated for model training.

The full dataset was generated by constructing negative interactions from unknown peptide–protein pairs through random pairing at a 1:1 ratio relative to the positive interactions. Experimentally validated non-interacting peptide–protein pairs are rarely available in public repositories; therefore, consistent with prior PepPI studies [27], unknown interaction pairs were treated as negative samples under a balanced random sampling scheme. Although random pairing may introduce relatively "easy" negatives, this approach reflects realistic biological data constraints in peptide–protein interaction modeling. To mitigate potential performance inflation, balanced sampling, k-fold cross-validation, and an independent hold-out test set were employed to ensure robust generalization assessment [47]. The overall dataset consisted of 6940 peptide–protein interaction pairs.

For the prediction data, cancer-related proteins were retrieved from UniProt [44] using cancer-process annotations; after curation, 895 proteins were stored in FASTA format. The C. *rhodostoma* venom-peptide dataset used in this study comprised 145 unique peptides in FASTA format and was obtained as secondary data from laboratory extractions studies conducted at Universitas Gadjah Mada.[1] Each peptide was paired with the 895 cancer-related proteins, yielding 129,775 peptide–protein combinations for interaction prediction using the best-performing trained model.

Feature extraction

Feature extraction was performed to encode peptide–protein pairs as fixed-length numerical vectors for input to deep-learning models. All sequences were padded to fixed lengths of 50 residues for peptides and 1,500 residues for proteins without altering the original sequence information. Two complementary representations were generated: descriptor-based and ProtT5 embedding-based features.

The first approach used four feature families: position-specific scoring matrix (PSSM) [48], amino acid sequence (AAS) [29], intrinsic disorder (ID) [49], and physicochemical properties (PP) [29]. PSSM features yielded 800 values (400 each for peptides and proteins), capturing residue conservation scores across sequence positions. AAS features produced 1550 values (50 for peptides and 1500 for proteins), encoding residue identities numerically. ID features produced 4650 values (150 for peptides and 4500 for proteins), reflecting per-residue disorder propensity. PP features contributed 1550 values (50 for peptides and 1500 for proteins), describing physicochemical attributes

[1] https://github.com/TropBRC-BioinfoLab/ProVenTL

such as polarity and hydrophobicity. Collectively, these four descriptors resulted in 8550 numerical features per peptide–protein pair.

We used ProtT5, a transformer-based protein language model pretrained on millions of sequences [50]. Its embeddings provided 1024 features per peptide and 1024 per protein (2,048 per pair). We used these embeddings to assess TL for improving PepPI accuracy.

ProVenTL

The ProVenTL (Fig. 2) is a deep-learning model for predicting interactions between C. *rhodostoma* venom peptides and human cancer proteins. It uses ProtT5 contextual embeddings pretrained on millions of protein sequences to generate biologically informative features for peptide–protein modeling. This design improves accuracy and training efficiency in data-limited settings. Embeddings from input sequences are extracted using ProtT5 and mean-pooled to a fixed-dimensional vector, as shown in Eq. (1).

$$\mathrm{x} = MeanPooling(ProtT5\,(s)) \tag{1}$$

The vector x is then passed to a stacked autoencoder (SAE) module comprising multiple hierarchical encoder layers [51]. Each encoder layer i applies a linear transformation followed by a nonlinear activation, as shown in Eq. (2).

$$z_{(i)} = f\left({W_{z(i)}}^{\top} z_{(i-1)} + b_{z(i)}\right), with z_{(0)} = x \tag{2}$$

where f denotes the nonlinear ReLU activation; dropout is applied before each encoder layer to mitigate overfitting; and W_z and b_z denote the weight matrix and bias vector of the $i = 1, 2, \ldots, k$ encoder layer, respectively. This procedure is repeated over k layers, yielding a latent vector $z_{(\mathrm{i})}$ that provides a compressed representation of the input. The encoder weights from each layer are stored and reused to initialize the subsequent deep neural network (DNN).

Binary classification is then performed using a DNN [52]. The latent vector $z_{(\mathrm{k})}$ is a fully connected network with a number of hidden layers matching the number of SAE encoder [53]. Each hidden layer applies ReLU, batch normalization, and dropout. The DNN forward pass is formulated as shown in Eq. (3).

$$\begin{aligned}
&h_{(1)} = f\left(W_{h(1)} z_{(0)} + b_{h(1)}\right)\\
&h_{(2)} = f\left(W_{h(2)} z_{(1)} + b_{h(2)}\right)\\
&\vdots\\
&h_{(L)} = f\left(W_{h(L)} z_{(k)} + b_{h(L)}\right)\\
&\hat{y} = \sigma(W_{\top} h_{(L)} + b)
\end{aligned} \tag{3}$$

where L denotes the number of hidden layers, f denotes the ReLU activation, σ denotes the sigmoid activation, and $\mathrm{W_h}$ and $\mathrm{b_h}$ denote the weight matrix and bias vector of the i

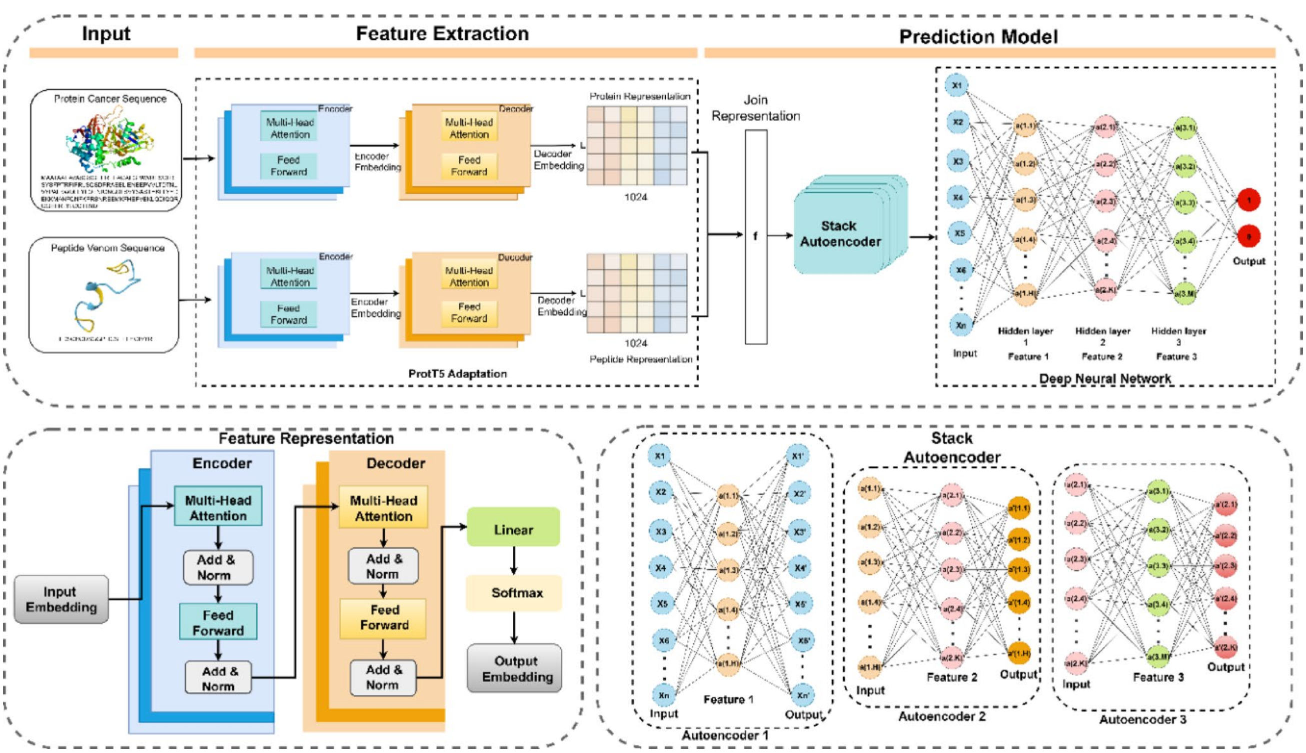


Fig. 2 The architecture of ProVenTL

hidden layer, respectively. The output $\hat{y} \in [0,1]$ indicates the probability that a peptide–protein pair interacts. The model is trained using the binary cross entropy loss, as shown in Eq. (4).

$$L_{BCE}(y,\hat{y}) = -\left[ylog\left(\hat{y}\right) + (1-y)\,log(1-\hat{y})\right] \quad (4)$$

where $y \in [0,1]$ denotes the ground-truth interaction label. Optimization is performed with the Adam optimizer, updating the weights based on error relative to the ground truth. After training, all network parameters $\theta = \{W, b\}$ are saved to an HDF5-formatted model, as shown in Eq. (5).

$$ProVenTL.h5 \leftarrow \theta \quad (5)$$

The trained model can perform inference directly on new peptide–protein pairs x_{new} without retraining, as shown in Eq. (6).

$$\hat{y}_{new} = ProVenTL\left(x_{new};\theta\right), \theta \in ProVenTL.h5 \quad (6)$$

By leveraging TL with ProtT5 embeddings and an SAE–DNN architecture, ProVenTL enables efficient prediction of biological interactions in data-limited domains. This approach supports in silico screening of snake-venom peptides for anticancer therapy development. The ProVenTL architecture is shown in Fig. 2.

Experimental setup and evaluation scenario

We used TL and embedding-based DL to predict interactions between C. *rhodostoma* venom peptides and cancer-related proteins. Two approaches were evaluated: a descriptor-based CAMP model and a ProtT5 embedding-based approach, focusing on ProVenTL (SAE–DNN) with TabNet as an additional comparator. For CAMP, peptide-protein descriptors (PSSM, AAS, ID, and PP) produced 8550 features per pair. We trained from scratch and TL (CAMP-TL), freezing embedding and convolutional layers and retraining only the final dense layer.

In the second approach, ProtT5 embeddings were used to represent peptide and protein sequences as contextual numerical vectors. These embeddings were used in two models: ProVenTL, which applies an SAE–DNN architecture comprising autoencoder pretraining followed by a supervised classifier, and TabNet, an interpretable model that uses sparse feature selection with attention-based decision steps. Both models were evaluated using (1) TL with ProtT5 embeddings and (2) from scratch training using conventional descriptor features (no pretraining). Accordingly, six modeling scenarios were examined in total, as listed in Table 1. These combinations of input features and training strategies were designed to assess each component's contribution to prediction accuracy and generalization.

Table 1 Predictive modelling scenarios

No	Scenario	Prediction Model	Feature Extraction
1	Model 1	CAMP from scratch	PSSM, ID, PP, AAS
2	Model 2	CAMP TL	PSSM, ID, PP, AAS
3	Model 3	ProVenTL	ProtT5
4	Model 4	SAE-DNN Descriptor	PSSM, ID, PP, AAS
5	Model 5	TabNet TL	ProtT5
6	Model 6	TabNet Descriptor	PSSM, ID, PP, AAS

Table 2 List of hyperparameter configurations

No	Algorithm	Hyperparameter	Values
1	CAMP from scratch	Learning rate	0.001, 0.0001
		Batch size	32, 64
		epochs	100, 200
2	CAMP TL	Learning rate	0.001, 0.01
		Batch size	32, 64
		epochs	50, 100
3	ProVenTL	Learning rate	0.0001, 0.001
	SAE-DNN Descriptor	Batch size	32, 64
		epochs	100, 200
4	TabNet TL	Mask_type	Sparsemax, entmax
	TabNet Descriptor	n_da/n_a	16, 32
		n_steps	3, 5
		Gamma	1.0, 1.5
		Learning rate	0.001, 0.005

The curated dataset comprised 6940 peptide–protein interaction pairs. Prior to model training, the dataset was divided into 80% training data (5552 pairs) and 20% independent test data (1388 pairs). Five-fold cross-validation was applied exclusively to the training set for internal model selection and hyperparameter optimization. The held out 20% test set was reserved for final external performance evaluation to assess generalization ability on unseen data. Hyperparameters were optimized using grid search, selecting the combination with the highest mean ROC–AUC across folds. Detailed hyperparameter settings for each model are listed in Table 2.

The list of hyperparameter configurations for each model is presented in Table 2. Learning rate controlled the magnitude of weight updates during backpropagation, thereby influencing convergence speed and training stability. Batch size, defined as the number of samples processed per update, affected gradient stability and generalization performance. Training epochs determined the overall duration of the learning process, ensuring sufficient pattern extraction while avoiding underfitting or overfitting [54].

For the TabNet models, mask_type specified the attentive feature selection mechanism. Model capacity was determined by the dimensionality of the decision and attention representations (*n_d* and *n_a*). Sequential decision refinement was governed by n_steps, whereas *gamma* regulated feature exploration and reuse across steps. The configuration yielding the highest mean ROC–AUC across cross-validation folds was selected as optimal for each model [35].

The best-performing model from each approach was subsequently applied to predict interactions between 145 unique *C. rhodostoma* venom peptides and 895 curated cancer-related proteins, generating 129,775 peptide–protein combinations for large-scale screening. Model performance was first assessed during cross-validation using standard classification metrics. Final predictions were filtered using a probability threshold ≥ 0.95, and statistical significance was determined based on score distribution-derived p-values [55], with pairs satisfying $p < 0.05$ prioritized as significant candidates [56].

Computational resources

Experiments were conducted on two on-premises systems at the Advanced Research Laboratory (ARLab), IPB University. The primary system was an NVIDIA DGX A100 under a Class-B ("medium") allocation, provisioned with four CPU cores and 24 GB of system RAM. The GPU was configured as a MIG 2 g.20gb slice (20 GB HBM2e) with exclusive access. Additional runs were performed on the Computer Science G6 HPC server equipped with an NVIDIA GeForce RTX 3060 (12 GB VRAM; 112 Tensor Cores). All experiments used Python 3.10 with NumPy, pandas, scikit-learn, TensorFlow, and PyTorch. CUDA and cuDNN versions matched the installed NVIDIA driver version. Random seeds were fixed across experiments, and identical preprocessing and evaluation pipelines were applied to ensure reproducibility.

Model evaluation

All models were evaluated using fivefold cross-validation to assess predictive performance and ensure consistency across folds. Although LOOCV is often applied to small datasets, it may yield overly optimistic estimates in pair-input prediction settings because of residual similarity between training and validation instances. Likewise, reliance on a single train–test split can lead to biased performance estimates [47]. Considering the moderate dataset size (6,940 interaction pairs) and the availability of an independent hold-out test set, fivefold cross-validation combined with external evaluation was adopted in accordance with established validation recommendations. This strategy mitigates performance inflation and provides a more reliable estimate of generalization performance [47].

Performance was assessed using five metrics: accuracy, precision, recall, F1-score, and area under the receiver operating characteristic (ROC) curve (AUC) [57], to quantify discrimination between interacting and non-interacting peptide–protein pairs. Accuracy measures the proportion of peptide–protein pairs correctly classified as interacting or non-interacting, as shown in Eq. (7).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Subsequently, precision quantifies the proportion of predicted positive cases that are actual interactions. This metric is critical for assessing the model's ability to identify true positive interactions, as shown in Eq. (8).

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

Recall quantifies the model's ability to identify actual positive interactions, measuring the proportion of true positives among all actual positive instances, as shown in Eq. (9).

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

To balance precision and recall, the F1 score harmonic mean is used, as shown in Eq. (10).

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{10}$$

The notation used in the above equations is as follows: TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives. Each model was compared using ROC curves and precision–recall curves (PRC) to identify the best-performing model. Higher recall reflects improved detection of positive interaction pairs [58].

Enrichment analysis

Using the best-performing model's predictions for interactions between C. *rhodostoma* venom peptides and cancer-associated proteins, we conducted a follow-up analysis on proteins predicted to have positive interactions. This analysis assessed the biological involvement of the target proteins and identified key pathways potentially affected by peptide interactions. Enrichment was performed with Enrichr [59], which provided access to functional resources, including Gene Ontology (GO) and the Kyoto Encyclopedia of Genes and Genomes (KEGG). The analysis covered the three GO domains, such as biological process (BP), molecular function (MF), and cellular component (CC) as well as KEGG pathways to gain deeper insight into the biological roles of the target proteins. The enrichment analysis was accompanied by a literature review for each finding to strengthen validity and assess its consistency with existing scientific knowledge.

Enrichment results were visualized as networks in Cytoscape v3.10.2 [60]. The visualizations showed relationships among peptides, predicted protein targets, and significantly enriched biological pathways, thereby facilitating functional interpretation of model predictions. Networks represented biological entities as nodes and edges, enabling more informative exploration of pathway–target interactions.

All procedures for feature extraction, model construction, hyperparameter selection, and evaluation are described above. The six modelling scenarios were systematically tested to assess classification performance for peptide–protein interaction prediction. The resulting performance comparisons, evaluation-metric visualizations, and biological interpretations are presented in section Result.

Molecular docking analysis

The three-dimensional coordinates of the target cancer-related proteins, PLAU (PDB ID: 1W0Z), PCSK9 (2P4E), HIF1AN (7A1J), CD274/PD-L1 (5N2F), and TRBC2 (4UDT), were retrieved from the RCSB Protein Data Bank (PDB) [61]. Structural preprocessing was performed in UCSF ChimeraX [62], including removal of crystallographic water molecules and isolation of biologically active polypeptide chains to establish a clean docking environment. Conformational modeling for the snake venom–derived peptides was performed using the PEP-FOLD 4 prediction server [63]. Molecular docking simulations were then performed using HPEPDOCK 2 [64, 65], with binding sites predicted using PrankWeb [66]. Peptide-protein complexes were evaluated and ranked according to their lowest docking energy scores. The physicochemical properties of the binding interfaces, including hydrogen bonding, hydrophobic contacts, and salt bridges, were characterized using the Protein–Ligand Interaction Profiler (PLIP) [67]. All molecular graphics and structural visualizations were rendered using UCSF ChimeraX [62].

Results

The predictive models were evaluated using fivefold cross-validation and five metrics, including accuracy, precision, recall, F1-score, and ROC–AUC. Six modeling scenarios were considered, combining three deep-learning architectures, CAMP, SAE–DNN, and TabNet, with two training strategies: from scratch and TL. Each model used one of two input feature types: protein descriptors (PSSM, ID, PP, and AAS) or ProtT5 embeddings. All models were tuned using grid search to select optimal hyperparameter configurations. Mean performance and training time for each model are listed in Table 3.

A comparative summary of model performance across the six scenarios is presented in Table 3. The table shows that the CAMP model trained from scratch yielded the lowest performance (accuracy 0.5292; ROC–AUC 0.5775). Despite 200 epochs with a learning rate of 0.0001 and a batch size of 32, it failed to effectively distinguish positives from negatives, as reflected by low recall (0.5779) and F1 score (0.5103). By contrast, the CAMP-TL model substantially improved performance, achieving an accuracy of 0.6754 and a ROC–AUC of 0.7300, despite training for only 100 epochs while fine-tuning only the final dense layer. These results indicate that TL effectively enhanced feature representations from the source to the target domain.

The ProVenTL model, which integrates ProtT5 embeddings with an SAE–DNN under a transfer-learning strategy, achieved the highest overall performance among all scenarios. It reached an accuracy of 0.7749 and a ROC–AUC of 0.8538, optimized using grid search (learning rate = 0.001, batch size = 32, epochs = 200). Its high precision (0.7911) and recall (0.7470) indicate strong positive-interaction detection with minimal classification error. As a baseline, the descriptor-based SAE–DNN trained with the same configuration performed slightly lower (accuracy = 0.7460; ROC–AUC = 0.8167), suggesting that ProtT5 representations and pretraining substantially improved feature quality and generalizability.

The TabNet–TL model, which also used ProtT5 embeddings, achieved an accuracy of 0.7576 and a ROC–AUC of 0.8378, demonstrating efficiency on high-dimensional, complex data with optimal hyperparameters (n_da = 32, n_steps = 3, gamma = 1.0, mask_type = entmax). As a baseline,

Table 3 Average fivefold evaluation metrics for peptide–protein interaction models

No	Scenario	Prediction Model	Accuracy	Precision	Recall	F1 score	ROC–AUC
1	Model 1	CAMP from scratch	0.5292 ± 0.0463	0.5560 ± 0.0657	0.5779 ± 0.3329	0.5103 ± 0.1646	0.5775 ± 0.0330
2	Model 2	CAMP TL	0.6754 ± 0.0276	0.6912 ± 0.0628	0.6638 ± 0.1258	0.6646 ± 0.0643	0.7300 ± 0.0442
3	Model 3	ProVenTL (Our proposed model)	**0.7749 ± 0.0155**	**0.7911 ± 0.0218**	**0.7470 ± 0.0447**	**0.7470 ± 0.0447**	**0.8538 ± 0.0195**
4	Model 4	SAE-DNN Descriptor	0.7460 ± 0.0181	0.7724 ± 0.0307	0.6982 ± 0.0383	0.7327 ± 0.0242	0.8167 ± 0.0180
5	Model 5	TabNet-TL	0.7576 ± 0.0183	0.7580 ± 0.0251	0.7551 ± 0.0265	0.7565 ± 0.0246	0.8378 ± 0.0173
6	Model 6	TabNet Descriptor	0.7172 ± 0.0131	0.7287 ± 0.0284	0.6945 ± 0.0347	0.7102 ± 0.0168	0.7919 ± 0.0191

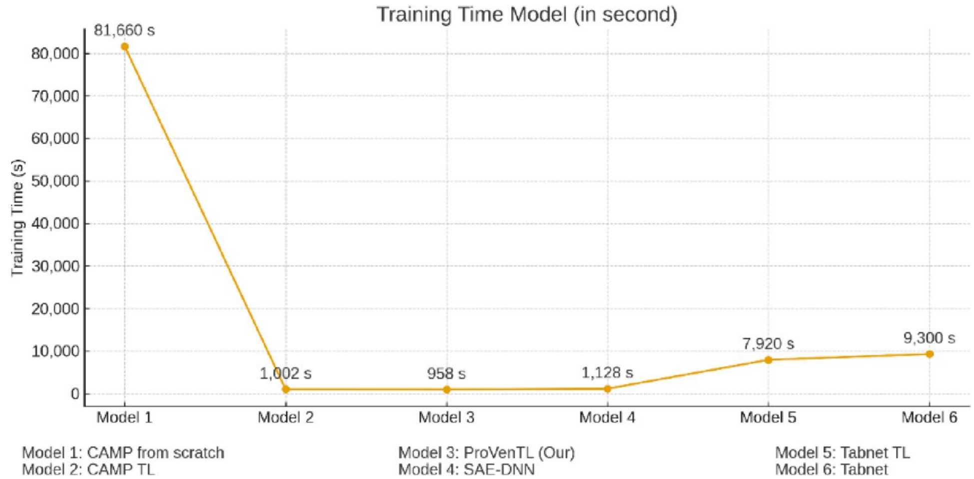


Fig. 3 Training time model

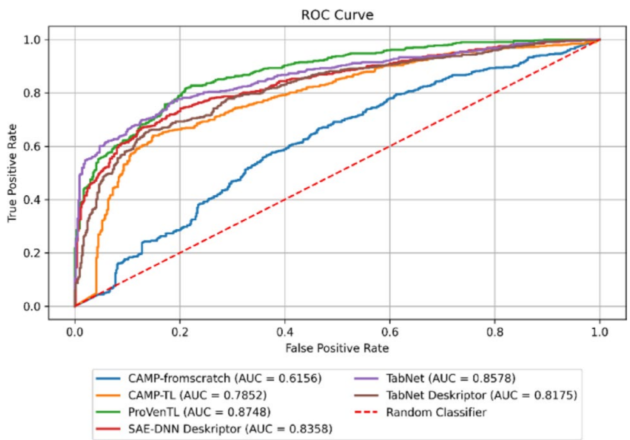


Fig. 4 Receiver operating characteristic (ROC) curves for the best validation fold of each predictive model

the descriptor-based TabNet model performed lower (accuracy = 0.7172; ROC–AUC = 0.7919), though it outperformed the CAMP model trained from scratch. These results reinforce the importance of embedding-based feature representations and transfer-learning strategies for improved classification performance. Training time efficiency is also crucial. As shown in Fig. 3, CAMP (from scratch) required more than 22 h owing to its fully trained convolutional architecture. In contrast, ProVenTL using ProtT5 embeddings with pretraining and fine-tuning reached optimal performance in 15 min 58 s. This demonstrates the computational efficiency of embedding-based TL while maintaining high predictive performance and emphasizes the importance of contextual representations and pretraining. We also present ROC and PR curves to assess performance across thresholds.

The ROC curves illustrated the classification performance of all predictive models, as shown in Fig. 4. These curves correspond to the best-performing validation fold for each model, whereas Table 3 reports the mean ROC–AUC across fivefold cross-validation. ProVenTL achieved the highest AUC of 0.8748, followed by TabNet–TL (0.8578) and SAE–DNN (descriptor) (0.8358), all substantially outperforming the random baseline (AUC ≈ 0.5). CAMP-TL attained an AUC of 0.7852, exceeding the from scratch CAMP model (0.6156) but trailing the embedding-based approaches. These results further underscore the contribution of ProtT5 representations and TL to improved classification accuracy.

The precision–recall (PR) curves for each model's best validation fold summarize overall classification performance, as shown in Fig. 5. ProVenTL consistently achieved the best results with an average precision (AP) of 0.8914, followed by TabNet–TL (0.8795) and SAE–DNN (descriptor) (0.8608). These three models maintained high precision across a wide range of recall, indicating robustness in identifying positive interactions. By contrast, the CAMP from scratch model exhibited a sharp precision drop as recall increased, yielding a substantially lower AP of 0.5651. These PR results further highlight the value of ProtT5-based embeddings and pretraining strategies for improving classification accuracy, particularly when training data are limited relative to model complexity.

For the performance assessment, a confusion-matrix analysis was conducted using each model's predictions on the test set to characterize error types when distinguishing interacting from non-interacting peptide–protein pairs. By examining correct predictions (TPs and TNs) and errors (FPs and FNs), we thoroughly evaluated each model's ability to detect peptide–protein interactions consistently. A summary of the confusion matrices for all six models is listed in Table 4.

An evaluation of the test-set confusion matrices in Table 4 indicates that the SAE–DNN (descriptor) achieved the most TPs (542), indicating strong sensitivity. ProVenTL achieved the most TNs (559) and the fewest FPs (135), reflecting high specificity/precision. TabNet–TL was well balanced, while CAMP from scratch had maximum errors

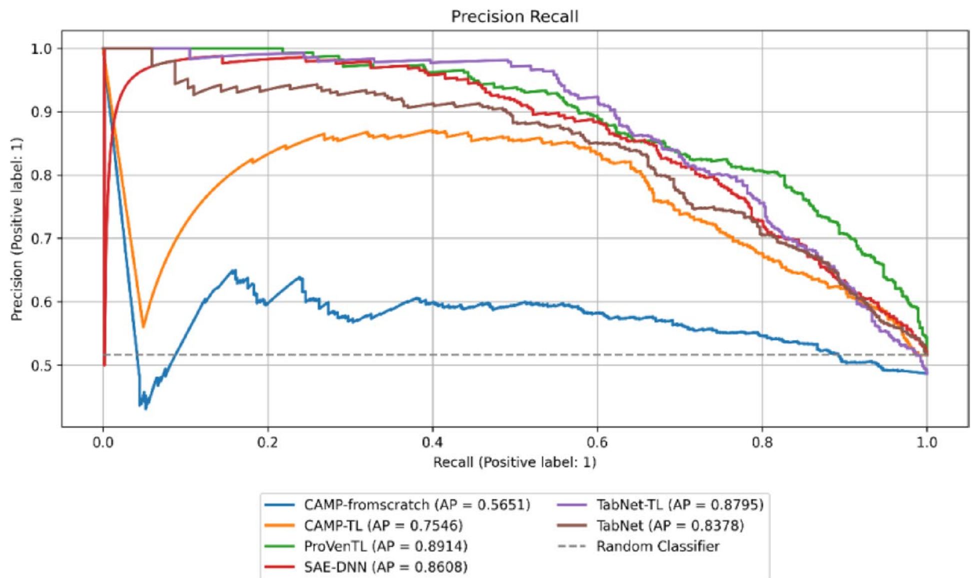


Fig. 5 Precision–recall (PR) curves for the best validation fold of each predictive model

Table 4 Confusion matrices based on test-set results

Matrix	Model					
	CAMP from scratch	CAMP TL	ProVenTL	SAE-DNN Descriptor	TabNet TL	TabNet Descriptor
True Positive (TP)	347	508	526	542	532	518
True Negative (TN)	430	462	559	498	548	470
False Negative (FN)	347	186	168	196	162	224
False Positive (FP)	264	232	135	152	146	176

(FN = 347; FP = 264), underscoring the limitations of training without embedding-based pretraining strategies.

A direct comparison with classical PepPI approaches further underscores the effectiveness of the proposed framework. The descriptor-based CAMP model, representing a conventional CNN-driven PepPI tool, achieved a ROC–AUC of 0.5775 when trained from scratch and 0.7300 under a transfer learning strategy. In contrast, ProVenTL attained a substantially higher ROC–AUC of 0.8538, reflecting an improvement of 0.1238 over CAMP-TL and 0.2763 over the from-scratch CAMP model. When compared with the descriptor-based SAE–DNN, which achieved a ROC–AUC of 0.8167, ProVenTL still demonstrated a performance gain of 0.0371, along with higher precision and a more balanced recall. Consistent improvements were also observed in the confusion-matrix analysis, where ProVenTL achieved the highest number of true negatives (559) and the lowest number of false positives (135), indicating superior discrimination between interacting and non-interacting pairs. These results indicate that integrating contextual ProtT5 embeddings with transfer learning confers consistent performance advantages over descriptor-based PepPI models.

Given these results, SAE–DNN with ProtT5 embeddings (ProVenTL) was selected as the primary model for large-scale interaction screening and subsequently applied to predict interactions between 145 *C. rhodostoma* peptides and 895 cancer-related proteins to obtain high-confidence candidates for biological and functional analysis.

Prediction of snake venom peptide cancer protein interactions

Interactions between 145 *C. rhodostoma* venom peptides and 895 curated cancer-related proteins were systematically screened using the best-tuned ProVenTL model, generating 129,775 peptide–protein combinations. These peptide–protein pairs were evaluated exclusively during the independent prediction phase and were not used for model training or validation. Predictions were filtered using a probability threshold ≥ 0.95, and statistical significance was determined based on score distribution-derived p-values, with pairs satisfying $p < 0.05$ retained as high-confidence candidates. From the 145 peptides in the independent prediction dataset, 45 peptides met these high-confidence criteria and were identified as participating in 498 statistically significant interactions involving 105 cancer-related proteins. The most frequently predicted targets included TRBC2 (27 peptides), CD274 (22), HIF1AN (21), PCSK9 (19), and PLAU (16). Table 5 lists the top ten peptide–protein pairs with the highest predicted interaction probabilities.

Overall, the prediction results indicate that ProVenTL not only produced accurate classifications on the training and validation sets but also identified high-confidence peptide–cancer protein interaction candidates. These findings provide a solid basis for downstream analyses, such as pathway enrichment, which are discussed in the following subsection.

GO and KEGG enrichment results

Enrichment analysis was performed on 105 cancer-related proteins identified in high-confidence interactions with 45 peptides, which represent a filtered subset of the 145 *C. rhodostoma* peptides screened in the independent prediction dataset. The proteins were analyzed using Enrichr across GO (BP/CC/MF) and KEGG databases. For GO-BP, the top ten significantly enriched terms (lowest *p*-values) were ranked by $-\log_{10}$ (*p* value). Overall, the top-ranked GO-BP terms concentrate on transporter regulation, plasminogen activation, and wound-response mechanisms, as shown in Fig. 6a, which details processes including negative regulation of sodium ion transmembrane transporter activity (GO:2,000,650), negative regulation of lipoprotein particle clearance (GO:0010985), negative regulation of plasminogen activation (GO:0010757), and regulation of response to wounding (GO:1,903,034). Additional terms involve receptor regulation and vasculogenesis, such as negative regulation of receptor binding (GO:1,900,121), positive regulation of receptor catabolic process (GO:2,000,646), and positive regulation of vasculogenesis (GO:2,001,214). These findings suggest functional clustering in ion transport, lipid metabolism, proteolysis, and vascular remodeling processes, which are commonly implicated in tumor progression and microenvironment adaptation, consistent with [68], who linked extracellular proteolysis and vascular remodeling to invasion and metastasis.

Network-level analysis of GO-BP identifies central hubs among the predicted proteins, as shown in Fig. 6b, indicating that PCSK9, CD274 (PD-L1), HIF1AN, and PLAU are highly connected to multiple enriched GO terms. PCSK9 is linked to ion transporter regulation and lipid metabolism, consistent with its role in cancer progression [69]. CD274 (PD-L1) is associated with receptor-binding and immune signaling pathways, reinforcing its function in tumor immune evasion and immune modulation within the tumor microenvironment [70]. HIF1AN connects to vasculogenesis and hypoxia-responsive processes, consistent with its role in cellular adaptation to low-oxygen conditions [71]. PLAU correlates with plasminogen activation and proteolytic regulation, supporting its involvement in extracellular matrix degradation and invasion [68]. Overall, these enrichment patterns provide a functional context for interpreting the predicted peptide–protein interactions, particularly in processes related to lipid metabolism, proteolysis, vascular remodeling, and immune regulation.

We used Enrichr to analyze GO-CC localization of proteins predicted to interact with C. *rhodostoma* peptides. Forty-two terms were identified; the top ten (lowest *p*-values) appeared in Fig. 7a based on $-\log_{10}$(*p*-value) and protein counts. These include *serine protease inhibitor complex* (GO:0097180), *early endosome* (GO:0005769), *serine-type endopeptidase complex* (GO:1,905,370), *endosome membrane* (GO:0010008), and *recycling endosome membrane* (GO:0055038). These terms indicate strong enrichment

Table 5 Top ten peptide–protein pairs with the highest predicted interaction probabilities

No	PDB Chain	Snake Venom Protein	Cancer Protein Symbol	Cancer Protein	Predicted Probability	z-score	*p* value
1	P30403_pep_9	Zinc metalloproteinase/disintegrin	PLAU	Urokinase-type plasminogen activator	0.9898	2.292117	0.010949
2	P30403_pep_9	Zinc metalloproteinase/disintegrin	PCSK9	Proprotein convertase subtilisin/kexin type 9	0.9888	2.246529	0.012335
3	P0CB14_pep_7	Snake venom metalloproteinase kistomin	HIF1AN	Hypoxia-inducible factor 1-alpha inhibitor	0.9883	2.226998	0.012974
4	P47797_pep_1	Thrombin-like enzyme ancrod-2	CD274	Programmed cell death 1 ligand 1	0.9881	2.216665	0.013323
5	P26324_pep_3	Thrombin-like enzyme ancrod	TRBC2	T cell receptor beta constant 2	0.9880	2.212999	0.013449
6	P30403_pep_9	Zinc metalloproteinase/disintegrin	HIF1AN	Hypoxia-inducible factor 1-alpha inhibitor	0.9878	2.206397	0.013678
7	P0CB14_pep_7	Snake venom metalloproteinase kistomin	CD274	Programmed cell death 1 ligand 1	0.9878	2.205937	0.013694
8	P30403_pep_9	Zinc metalloproteinase/disintegrin	CD274	Programmed cell death 1 ligand 1	0.9876	2.196986	0.014011
9	Q9I841_pep_1	Snaclec rhodocytin subunit alpha	HIF1AN	Hypoxia-inducible factor 1-alpha inhibitor	0.9871	2.17497	0.014816
10	P26324_pep_1	Thrombin-like enzyme ancrod	HIF1AN	Hypoxia-inducible factor 1-alpha inhibitor	0.9870	2.171432	0.014949

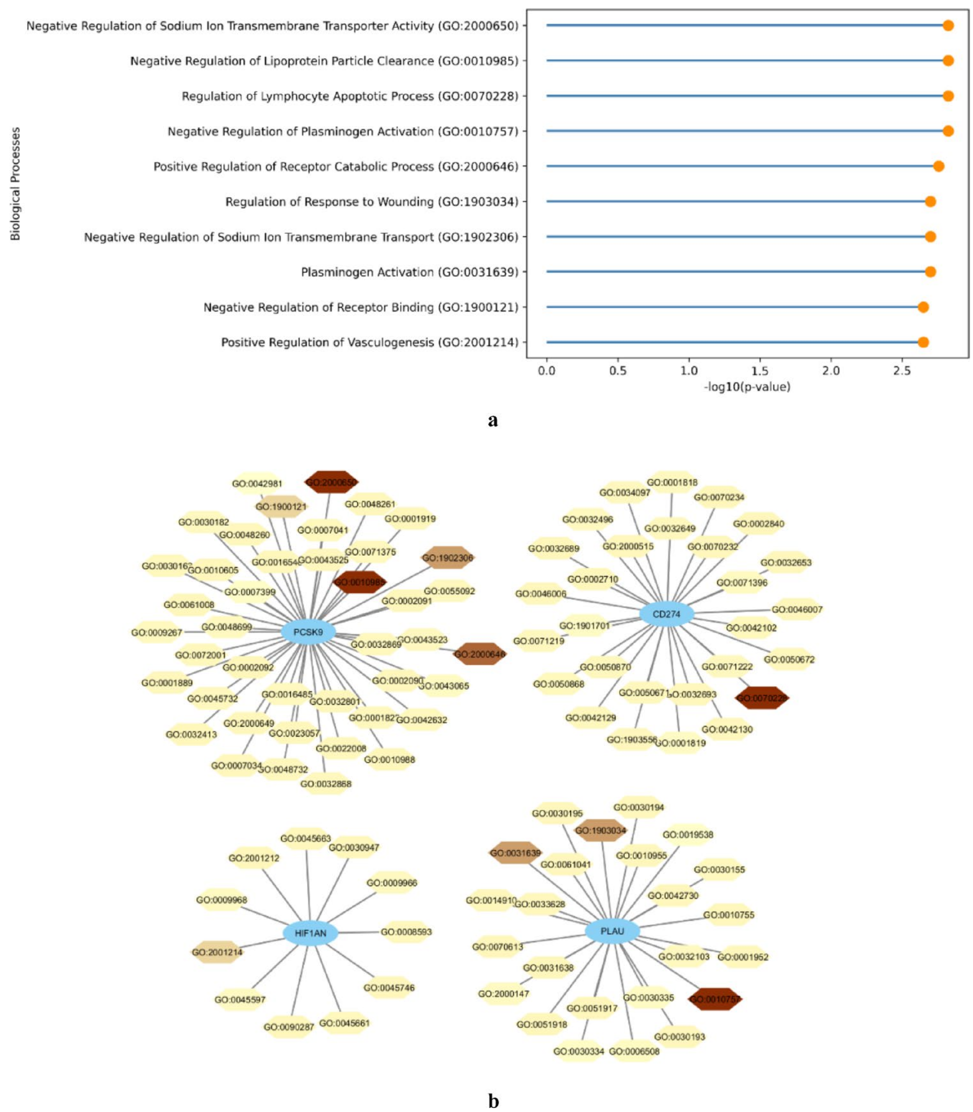


Fig. 6 Gene Ontology Biological Processes (GO-BP): **a** top 10 terms with the lowest p values; **b** protein GO process network, where darker colors indicate more significant terms ($p<0.01$)

in vesicular trafficking systems and protease-related complexes, which are central to proteolysis, membrane protein recycling, and intracellular transport. The dominance of endosome-and endolysosome-associated components suggests involvement of the predicted proteins in endocytic pathways and lysosomal degradation mechanisms that regulate receptor turnover and cellular signaling, consistent with [72], which described the essential role of endosomal trafficking in controlling receptor activity and signal transduction. In addition, enrichment of the serine protease inhibitor complex highlights regulatory control over proteolytic activity, a process closely linked to extracellular matrix remodeling, invasion, and metastatic progression, as supported by [73], who demonstrated that dysregulated protease activity contributes to tumor dissemination and tissue remodeling.

Topological analysis of the GO-CC network identifies central proteins within endosomal membranes and protease-related complexes, as shown in Fig. 7b. In this network, CD274, PCSK9, PLAU, and HIF1AN emerge as key hubs. CD274 (PD-L1) is associated with the endosome and recycling endosome membrane, supporting its role in membrane recycling and immune regulation, consistent with [74]. PLAU is linked to the serine-type endopeptidase complex and the serine protease inhibitor complex, reflecting its involvement in proteolytic activity and tissue remodeling. PCSK9 binds to the endosomal and endolysosomal membranes, in agreement with [69], which described its role in vesicular trafficking and receptor degradation. HIF1AN is associated with cytoplasmic protein complexes, consistent with its regulatory function in hypoxia response. These interactions highlight coordinated roles in vesicle trafficking, proteolysis, and hypoxia signaling, all of which are commonly implicated in immune regulation and tumor progression.

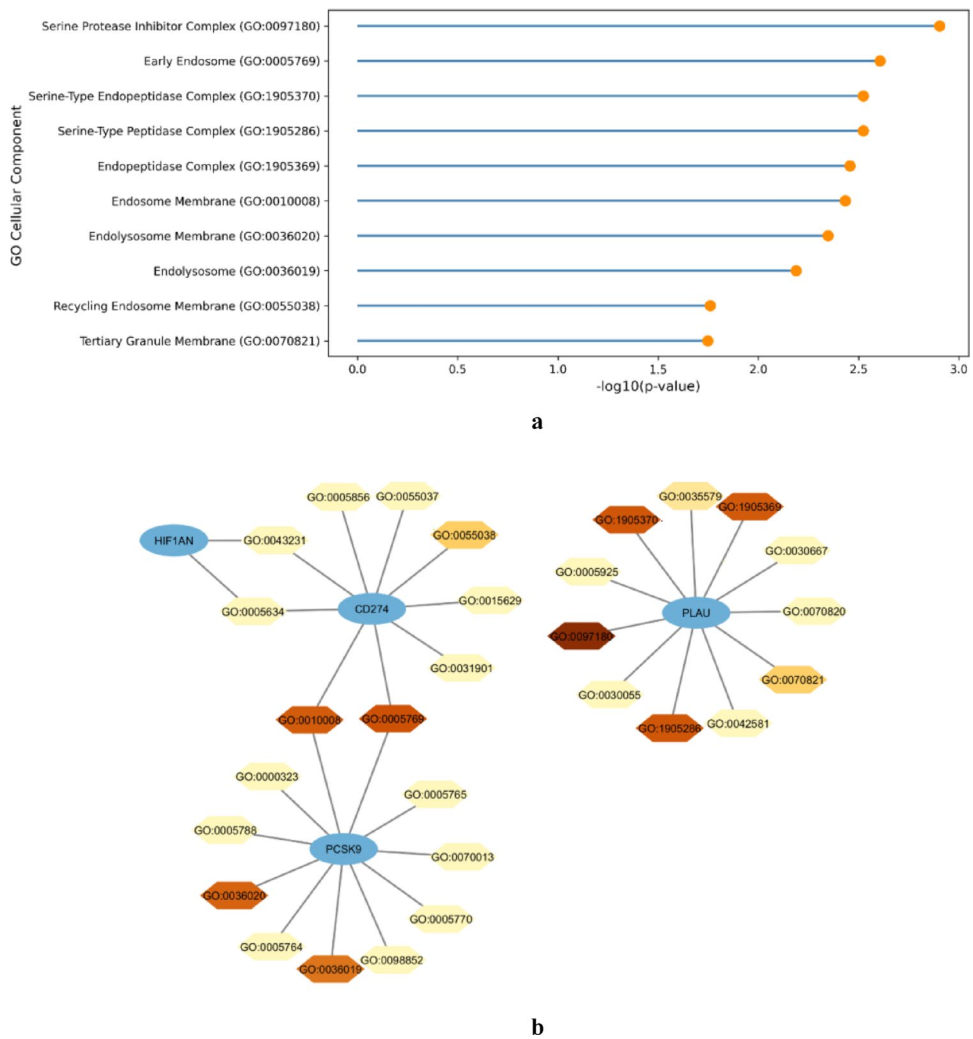
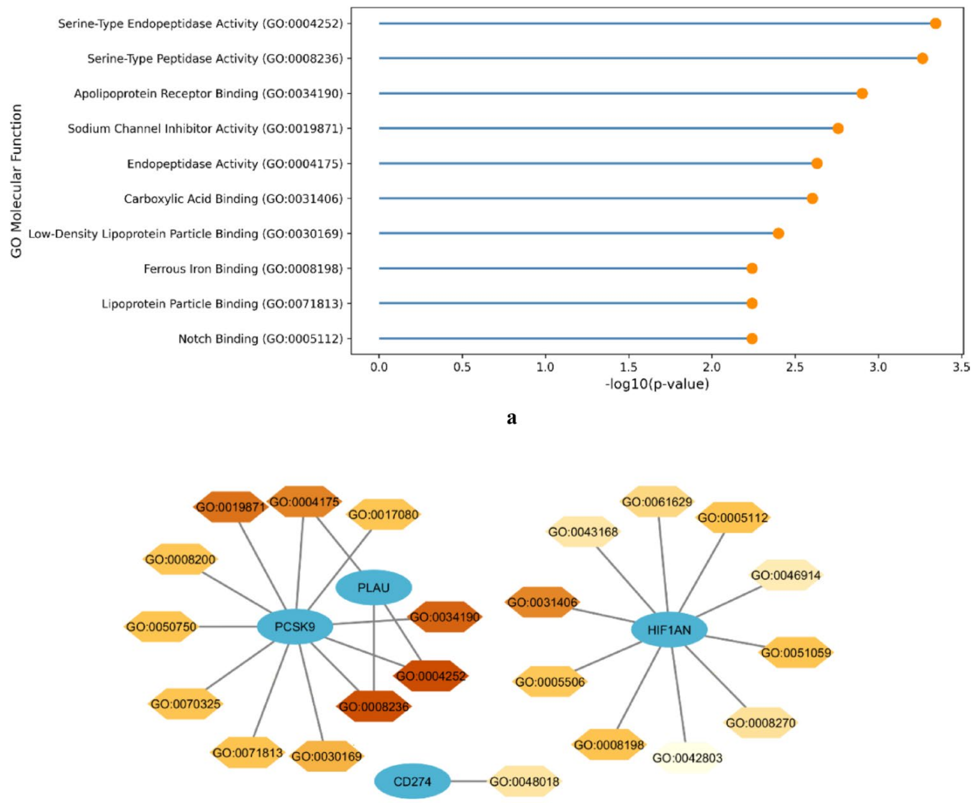


Fig. 7 Gene Ontology Cellular Component (GO-CC): **a** top 10 terms with the lowest *p* values; **b** protein GO-CC network, where darker colors indicate more significant terms ($p < 0.01$)

Molecular function enrichment identifies dominant serine-type proteolytic activities together with lipid-binding and transmembrane signaling functions. As shown in Fig. 8a, GO-MF enrichment highlights the top terms ranked by $-\log_{10}(\mathrm{p})$, including serine-type endopeptidase activity (GO:0004252), serine-type peptidase activity (GO:0008236), and endopeptidase activity (GO:0004175), indicating strong involvement of proteolysis [75]. Additional enriched terms such as apolipoprotein receptor binding (GO:0034190), lipoprotein particle binding (GO:0071813), sodium channel inhibitor activity (GO:0019871), and Notch binding (GO:0005112) suggest roles in lipid homeostasis and transmembrane signaling. These functions implicate coordinated regulation of proteolysis, metabolic transport, and cell communication, processes closely linked to tumor invasion and microenvironment adaptation [76].

Within the GO-MF network, several predicted targets serve as hubs. As shown in Fig. 8b, PCSK9 forms the primary hub within the protease–lipid functional cluster. At the same time, PLAU and CD274 share overlapping molecular functions within this network, including serine-type endopeptidase activity, serine-type peptidase activity, endopeptidase activity, and apolipoprotein receptor binding, highlighting combined roles in proteolysis and lipid interaction, consistent with the established metabolic regulatory functions of PCSK9. HIF1AN appears as a distinct functional cluster associated with molecular binding and protein interaction regulation, reflecting its established role in modulating hypoxia-inducible factor signaling and cellular adaptation to hypoxic stress, consistent with recent reviews on HIF-1–mediated tumor progression [77]. Node intensity (darker color indicating lower p-values) shows that protease- and lipid-binding–related functions linked to PCSK9 exhibit higher statistical significance than those associated with HIF1AN, underscoring the dominant contribution of proteolytic and lipid-regulatory activities within this network.

KEGG enrichment of the predicted targets prioritizes cancer-related signaling and immune modules, as shown in Fig. 9a. The top six pathways ranked by $-\log_{10}(p)$



b

Fig. 8 Gene Ontology Molecular Function (GO-MF): **a** top 10 terms with the lowest *p* values; **b** protein GO-MF network, where darker colors indicate more significant terms ($p < 0.01$)

include complement and coagulation cascades (hsa04610), Prostate cancer (hsa05215), NF-kappa B signaling pathway (hsa04064), Transcriptional misregulation in cancer (hsa05202), Proteoglycans in cancer (hsa05205), and MicroRNAs in cancer (hsa05206). Collectively, these pathways reflect integrated processes involving inflammation and immune activation (complement and NF-κB signaling), transcriptional dysregulation, extracellular matrix modulation, and miRNA-mediated regulation mechanisms frequently implicated in tumor initiation and progression, including immune signaling and tumor microenvironment modulation, consistent with previous reports [78]. These findings indicate that the predicted protein targets are enriched in biologically relevant oncogenic pathways.

In line with the KEGG results above, target-centric mapping highlights PLAU's extensive involvement across cancer pathways. As shown in Fig. 9b, PLAU is linked to Proteoglycans in cancer (hsa05205), MicroRNAs in cancer (hsa05206), Transcriptional misregulation in cancer (hsa05202), Prostate cancer (hsa05215), Complement and coagulation cascades (hsa04610), and NF-kappa B signaling pathway (hsa04064); darker nodes denote $p < 0.01$. These connections are consistent with the reported roles of PLAU in ECM remodeling and immune-related signaling processes within the tumor microenvironment [79]. TRBC2 was not detected in the KEGG enrichment, suggesting limited pathway annotation and the need for further validation.

Molecular docking analysis

Docking analysis was performed to evaluate the interaction potential of snake venom–derived peptides with cancer-related proteins. The binding strength of each peptide–protein complex was assessed using the HPEPDOCK docking score, which provides an estimate of the stability and favourability of the interaction [64, 65]. The results showed that all candidate peptides bound their respective targets with moderate to strong predicted affinity. The docking scores for all complexes and key interactions, including hydrogen bonds and hydrophobic contacts, are summarised in Table 6.

Control peptides were included to support docking validation. The Å6 peptide, a capped 8-amino-acid inhibitor derived from uPA, was used as a control for the uPA/uPAR (PLAU) system due to its reported downregulation of uPAR-mediated signaling [80]. A synthetic LDLR EGF-A domain peptide was used as a control for PCSK9 interactions [81], while a phosphorylated HIF-1α peptide (residues 788–806, Thr-796 phosphorylated) was used for comparison in HIF1AN docking [82]. For immune checkpoint validation, the PD-L1–derived L11 peptide was included

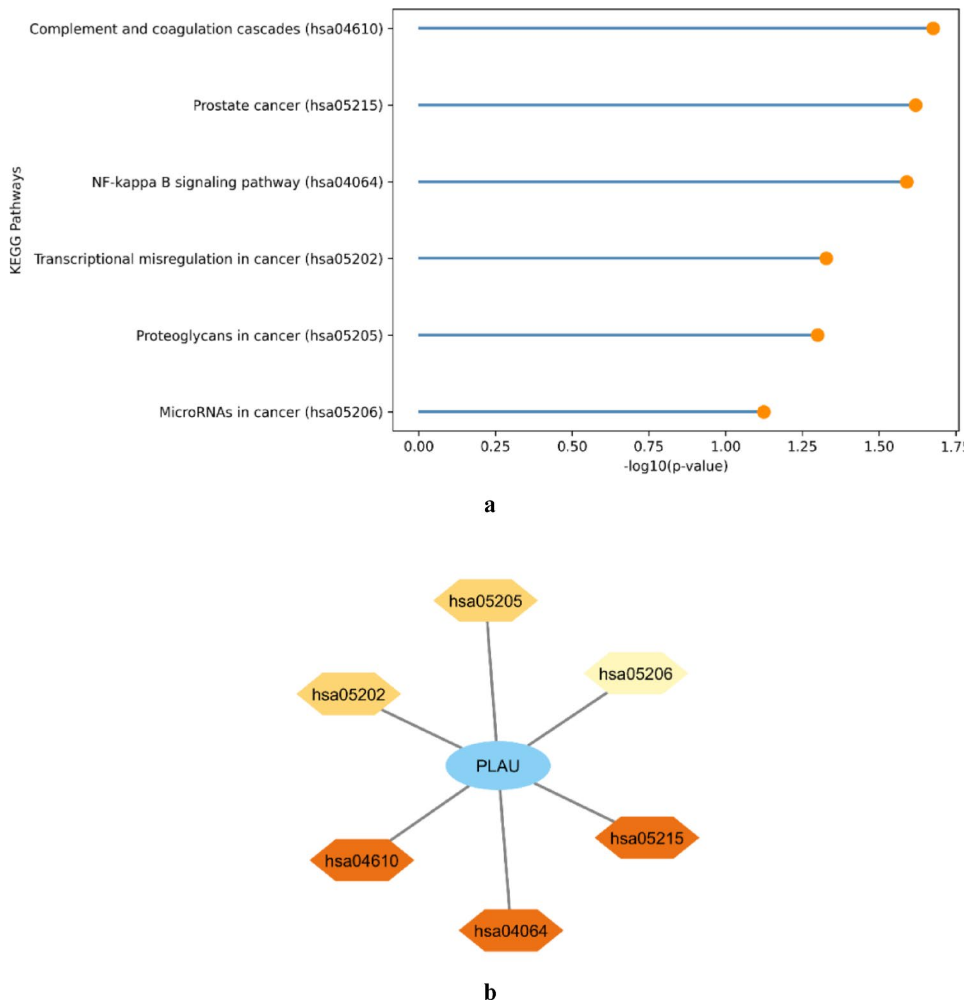


Fig. 9 KEGG pathways: **a** top 10 terms with the lowest p values; **b** protein KEGG pathway network, where darker colors indicate more significant terms ($p<0.01$)

as a PD-1 (CD274) binding control [83]. In addition, the TRBC2 core peptide (CP), derived from the T-cell receptor α-chain and reported to exhibit immunomodulatory activity in inflammatory disease models, was also included as a control ligand [84]. These peptides were used as reference compounds to evaluate docking interactions across the selected protein targets. As shown in Fig. 10.

In conjunction with the ProVenTL framework, peptide–protein docking analysis identified peptide–target pairs with varying predicted interaction probabilities. Based on the ProVenTL scoring output, the peptide–protein pair with the highest docking score was considered to have the highest predicted interaction probability. Among the tested peptides, P30403_pep_9 demonstrated the most favourable predicted interaction, with a docking score of −194.503, followed by the reference control peptide Å6 (KPSSPPEE) (−158.255), indicating comparatively stronger predicted binding propensity. Interaction analysis using PLIP showed that P30403_pep_9 formed multiple hydrophobic contacts with LEU97B and TRP215, along with additional non-polar interactions involving TYR60B, THR39, GLY216, PRO225, and VAL227. Hydrogen bonding interactions were also observed with LYS143 and ASP189, suggesting potential stabilisation within or near the functional region of PLAU.

Consistent with these findings, P30403_pep_9 also exhibited strong predicted binding tendencies toward other cancer-associated targets, particularly PLAU and HIF1AN. Similarly, P0CB14_pep_7 showed a favourable predicted affinity toward immune-related targets, including CD274. The interaction patterns were primarily driven by hydrophobic contacts, supported by hydrogen bonds, which are commonly associated with stable peptide–protein complexes [85, 86].

Discussion

We showed that TL with DL accurately predicted interactions between C. *rhodostoma* peptides and cancer proteins. Data scarcity remains a core obstacle, but ProtT5 embeddings with SAE–DNN fine-tuning (ProVenTL) yielded the top

Table 6 The docking scores for all complexes and key interactions, including hydrogen bonds and hydrophobic contacts

Cancer Protein	Peptide	Best Docking Score	Amino Acid Residue Interactions		
			Hydrophobic Interactions	Hydrogen Bonds	Salt Bridges
PLAU	P30403_pep_9	−194.503	LEU97B, TRP215	TYR60B, THR39, GLY216, PRO225, VAL227	LYS143, ASP189
	KPSSPPEE	−158.255	LEU97B	ARG37A, TYR60B, ARG35, THR39, GLN192, ARG217, GLY219	HIS37
PCSK9	P30403_pep_9	−146.736	VAL79, VAL81, LYS110, VAL114, PHE115, LEU119	ARG96, ILE111, HIS113, HIS116, ASP146	LYS110, HIS116, GLU144
	GTNECLDNNGGCSHVCNDLKIGYECLCPDGFQLVAQRRCEDI	−160.72	PHE115, LEU118	LYS110, HIS113, LEU118, ASP146	
HIF1AN	P0CB14_pep_7	−164.326	TRP296, ILE306, LEU310, LYS315, ALA317, ILE318, ASN321, ILE322	ASP89, SER91, ASP104, GLN148, ASN151, GLN203, ASN321	LYS106
	P30403_pep_9	−172.873	TYR102, GLU105, ASN119, GLN147, THR183, LEU186, ASP201, TRP196	ASP104, GLU105, PRO116, ARG120, ASP201	ASP104
	Q9I841_pep_1	−148.773	TYR102, LYS106, GLN147, THR183	SER91, TYR93, GLU122, ASN151, THR183, SER184	
	P26324_pep_1	−141.019	LYS298, THR302, ILE306, ALA317, ASN321	GLN148, ASN151, GLU202, GLN203, TYR276, ALA300, ASN321, LYS324	
	DESGLPQLTSYDCEVNAPIQGSRNLLQGEELLRAL	−113.46	THR183, LEU186, GLN203, TRP296	TYR102, LYS106, ARG120, GLN147, THR149, GLU202, GLN203, ARG238	

Table 6 (continued)

Cancer Protein	Peptide	Best Docking Score	Amino Acid Residue Interactions		
			Hydrophobic Interactions	Hydrogen Bonds	Salt Bridges
CD274	P47797_pep_1	−160.437	ILE54(B), TRY56(B), VAL68(B), ASP122(A), TYR123(A)	PHE19(A), ASN63(B), GLN66(B), ALA121(A)	
	P0CB14_pep_7	−175.555	GLN47(A), TYR56(B), VAL68(B), ASP122(A), TYR123(A)	PHE19(A), GLN47(A), ASN63(B), GLY119(A), TYR123(A), LYS124(A)	ASP49(A), ARG125(A)
	P30403_pep_9	−162.753	ILE54(B), TYR56(B)	ALA18(A), PHE19(A), PRO43(A), TYR56(B), ASN63(B), MET115(B), ALA121(A), ASP122(A), ARG125(A)	
	VCRXXISYGGADYKRCT	−174.940	ASN63, VAL76	ALA18, TYR56, ASN63, ARG125	ASP49, ASP73
TRBC2	P26324_pep_3	−166.813	PHE48(A), LYS55(A), ASN57(A), LEU97(B), TYR100(B), TYR103(B)	HIS37(A), ASN46(A), LEU47(A), ASN57(A), ARG59(A), THR81(A), TYR100(B), GLU101(B), TYR103(B), ARG165(A), LYS170(A)	GLU122
	GLRILLLKV	−161.791	ASN63, VAL76	ALA18, TYR56, ASN63, ARG125	ASP49, ASP73

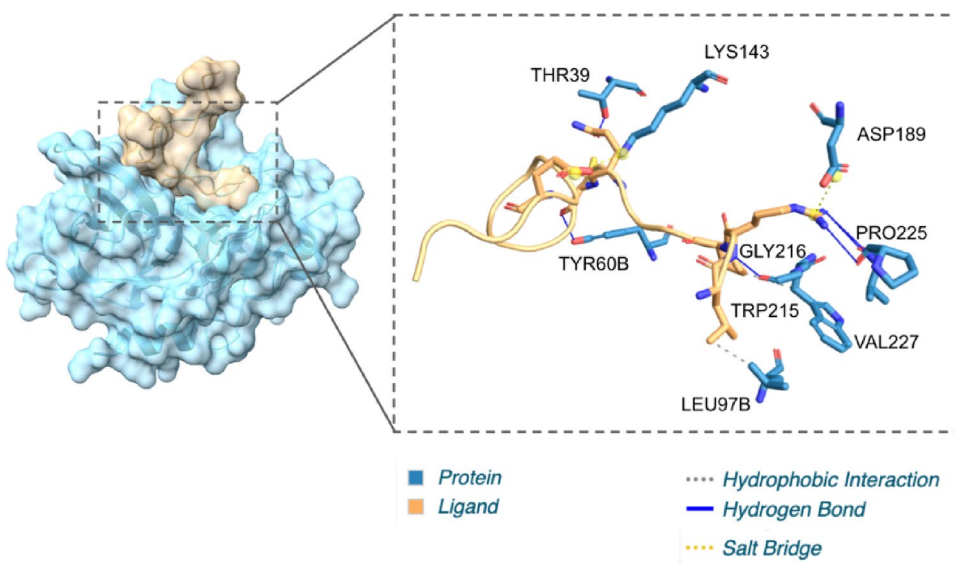


Fig. 10 Structural visualization and residue interaction profile of P30403_pep_9 in complex with PLAU

test scores (ROC–AUC 0.8626; accuracy 0.7817), outperforming from scratch and descriptor-based models (CAMP, TabNet). Two components underpinned ProVenTL's gains. First, ProtT5 embeddings, trained on billions of sequences with a Transformer, captured structural/functional context without MSA and outperformed conventional methods on secondary structure and localization [28, 50], consistent with prior studies reporting that transformer-based embeddings improve downstream protein tasks. Second, SAE learned latent features and reduced dimensionality, producing compact, stable representations that curbed the curse of dimensionality. Bypassing the SAE and feeding ProtT5 embeddings directly to a DNN increased overfitting, compute cost, and convergence time, whereas the SAE filtered signal from noise, improving efficiency in agreement with findings on viral classification and PPI prediction [87–89].

Post-encoding compressed latent features were fed to a DNN, the primary classifier for complex and nonlinear relationships. The SAE–DNN pairing accelerated training, improved efficiency, and yielded more stable, robust performance, particularly in data-limited PepPI settings, consistent with prior studies reporting improved stability and generalization for SAE–DNN pipelines [90, 91]. This synergy provided ProVenTL with higher accuracy and reduced overfitting compared with baselines.

By incorporating a TL scheme, models can be fine-tuned on limited target domains using pretrained parameters, thereby improving generalization while mitigating overfitting. This approach has been widely applied to drug-target interaction (DTI) prediction and has proven promising for addressing data scarcity by leveraging knowledge from broader, related source domains [32, 92]. In this study, all TL variants, such as CAMP-TL, TabNet-TL, and ProVenTL, achieved better evaluation metrics than their from scratch counterparts. These results support that transferring knowledge from a general domain (pretraining) to a specific domain (fine-tuning) enriches internal representations and acts as a regularizer, stabilizing training and enhancing generalization [30]. TL improved training efficiency, faster convergence, and fewer epochs by providing informative, stable initialization, and avoiding inefficient parameter-space exploration. This is advantageous under limited compute. TL models also showed more consistent cross-validation results, with smaller fold-to-fold fluctuations in ROC–AUC and accuracy, indicating stronger robustness to data variability.

ProVenTL (ProtT5 + SAE-DNN) predicted 498 high-confidence peptide–protein pairs (probabilities ≥ 0.95 and p values < 0.05) across 45 peptides and 105 cancer proteins**.** Importantly, examination of the structural classes represented among the top-ranked peptide–protein pairs (Table 5) indicates that the highest-confidence predictions are not randomly distributed across venom peptides but are predominantly derived from zinc metalloproteinases, disintegrin-related peptides, thrombin-like serine proteases, and C-type lectin–like proteins. Snake venom metalloproteinases (SVMPs) are $Zn2^{+}$ -dependent endopeptidases characterized by conserved catalytic domains and modular architectures that enable extracellular matrix (ECM) degradation and hemostatic regulation [93]. These enzymes, together with disintegrin-containing proteins, are organized into multidomain structures that facilitate modulation of integrin-mediated adhesion and related processes associated with angiogenesis and tumor progression [94]. Disintegrins possess conserved cysteine-rich scaffolds and integrin-binding motifs that enable binding to integrins and disruption of cell adhesion, thereby contributing to anti-angiogenic and anti-metastatic effects [95]. Recent reviews further emphasize that venom-derived peptides represent structurally stable and evolutionarily optimized scaffolds for cancer-targeted drug development [4, 96].

Therefore, the predominance of proteolytic and ECM-interacting venom families among the strongest ProVenTL predictions aligns with the enrichment of proteolysis, complement cascades, receptor signaling, and extracellular matrix–related pathways identified in the GO and KEGG analyses. This structural clustering supports the biological coherence of the predicted interactions and suggests that ProVenTL captures biologically meaningful patterns rather than arbitrary sequence correlations. Among the enriched immune-related targets, CD274 (PD-L1) emerged as a central hub, providing a representative example of how venom-derived proteolytic classes may intersect with immune checkpoint regulation. CD274 (PD-L1) is a key immune checkpoint protein that mediates tumor immune evasion by interacting with the PD-1 receptor on activated T cells, leading to suppression of cytotoxic activity and T-cell dysfunction [97]. At the genomic level, amplification of the 9p24.1 locus containing CD274 has been reported to correlate with increased cytolytic activity, suggesting an adaptive response of tumor cells to immune pressure within the tumor microenvironment [98]. Furthermore, PD-L1 expression is inducible by IFN-γ via the JAK/STAT signaling pathway, indicating that its regulation is influenced by inflammatory signals and local immune interactions [97].

To further illustrate this mechanism at the level of specific venom components, ancrod-2, a snake venom serine protease (SVSP), exhibits fibrinogenolytic activity and contributes to hemostatic regulation, with potential implications for extracellular matrix remodeling [96]. Extracellular matrix remodeling is known to regulate cytokine availability and influence immune cell infiltration within the tumor microenvironment [99, 100], thereby modulating inflammation-responsive signaling pathways associated with CD274 expression. Based on these findings, we hypothesize that the proteolytic activity of ancrod-2 may modulate CD274 expression by altering the microenvironment and influencing the IFN-γ/JAK/STAT pathway, and potentially through direct or indirect interactions that require further structural validation. This framework suggests a mechanistic link between enriched immune and extracellular matrix pathways and the biological activity of venom-derived proteases in cancer.

Beyond immune checkpoint regulation, additional predicted targets such as PCSK9 further support the involvement of ProVenTL-identified interactions in immune–metabolic pathways relevant to cancer progression. PCSK9 is best known for its role in lipid metabolism by promoting LDL receptor (LDLR) degradation. However, emerging evidence indicates that this protein is also involved in cancer progression and immune regulation. PCSK9 has been reported to inhibit apoptosis via by activating the FASN–Bax/Bcl-2–Caspase9/Caspase3 pathway and is associated with a poor clinical prognosis [101]. In addition, PCSK9 reduces MHC-I expression on tumor cells, thereby impairing CD8 + T-cell recognition and promoting immune evasion [102]. Its expression correlates with immune infiltration and activation of inflammatory pathways across multiple cancer types [103]. Interestingly, proteolytic enzymes such as MMP2 have been shown to neutralize PCSK9 and prevent LDLR degradation, suggesting potential crosstalk between proteolytic mechanisms and PCSK9-mediated oncogenic pathways [104].

In this context, snake venom–derived zinc metalloproteinases and disintegrins are $Zn2^{+}$-dependent proteases that degrade extracellular matrix (ECM) components and inhibit integrin-mediated signaling [105, 106]. ECM remodeling is known to modulate inflammatory pathways, such as NF-κB and STAT3, within the tumor microenvironment [100], which are also involved in regulating immune–metabolic mediators, including PCSK9 [103]. Based on these interconnected mechanisms, zinc metalloproteinase/disintegrin peptides may modulate PCSK9 function through tumor microenvironment remodeling and, potentially, through proteolytic interactions that require further experimental validation.

Limitations include the absence of in vitro or in vivo validation; therefore, predicted interactions remain computational hypotheses that require experimental confirmation. In addition, although transfer learning mitigated data scarcity, the overall dataset size remains relatively limited compared with large-scale protein–protein interaction corpora, potentially constraining model generalization to broader biological contexts. Potential sampling bias may also arise from the specific selection of venom-derived peptides and curated cancer-related proteins, which might not fully represent the diversity of peptide–protein interactions in vivo. Furthermore, negative samples in this study represent unknown interactions rather than experimentally confirmed non-binding pairs. Because these negative samples were generated through random pairing, relatively less challenging ("easy") negatives may be present, which could influence classification difficulty despite the use of balanced sampling and independent validation procedures. Accordingly, the proposed model is designed to distinguish known interacting pairs from putative unknown pairs and is not intended to serve as a definitive predictor of true biological non-interactions.

Future work should incorporate expanded and more diverse datasets, independent external validation cohorts, and systematic wet-lab verification to strengthen biological reliability. Further methodological improvements may include progressive unfreezing and adaptive learning rate schedules for more efficient fine-tuning; alternative embeddings (e.g., ESM-2, AlphaFold-derived features) to enhance biological fidelity; and interpretability analyses using SHAP and attention mechanisms to identify influential features. Active transfer learning could also prioritize informative subsets, improving computational efficiency and enabling staged experimental validation. These directions extend the exploration of C. rhodostoma peptides and reinforce the role of DL/TL as powerful tools for accurate, efficient, and interpretable peptide-based cancer therapy discovery.

Our results validate ProVenTL and motivate therapeutic exploration of natural peptides against enrichment-identified cancer pathways. They highlight the usefulness of deep-learning TL for peptide–protein interaction modeling and support snake-venom peptides as targeted therapeutics.

Conclusions

This study demonstrates that a DL-based TL approach effectively predicted interactions between natural peptides derived from the venom of C. *rhodostoma* and cancer-related proteins. Among six evaluated architectures, the SAE–DNN model using ProtT5 embeddings ProVenTL achieved the best test-set performance (ROC–AUC = 0.8626; accuracy = 0.7817). The model identified 498 significant peptide–protein interactions (prediction probability $\geq$ 0.95 and *p* value < 0.05), involving 45 unique peptides and 105 unique cancer-related proteins. Key targets, including TRBC2, CD274, HIF1AN, PCSK9, and PLAU, were enriched in pathways related to proteolysis, lipid metabolism, vesicular trafficking, immune signaling, and transcriptional dysregulation, highlighting the potential of the predicted peptides to modulate tumor-associated regulatory networks.

Results show that ProtT5 captured sequential and contextual signals, and TL improved generalization. ProVenTL is training-efficient and supports interpretability. A key limitation is the lack of in vitro and in vivo validation. Future study must include biological validation, SHAP/attention, and advanced embeddings (ESM-3, AlphaFold-derived) and apply active learning to select informative pairs. Overall, the study supports natural-peptide therapeutics and underscores TL computational frameworks in modern drug discovery.

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Data availability All source code, configuration files, and trained model checkpoints are available at https://github.com/TropBRC-BioinfoLab/ProVenTL

Declarations

Conflict of interest The Authors declare no conflict of interest.

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